Sanjana Joshi

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Portfolio: astroartics.github.io/Sanjana-Joshi-Portfolio

Education

Savitribai Phule Pune University
M.Sc. in Scientific Computing — 2024 - Present
First Year GPA: 9.14 / 10.0

Savitribai Phule Pune University
B.Sc. in Computer Science — March 2021 – April 2024
CGPA: 9.71 / 10.0

Projects

1. **Optimizing Minor Determinant Calculations in ECDLP** | Research Project — January 2025 - April 2025
   - A cryptography-based research project aimed at reducing computational complexity in matrix-based approaches to solving the Elliptic Curve Discrete Logarithm Problem (ECDLP).
   - Proposed a probabilistic pruning algorithm to eliminate higher-order minors unlikely to yield zero determinants, based on 2×2 minor patterns.
   - Gained familiarity with basic finite field operations (addition, multiplication, inverses) and implemented them over a 15-bit prime field as part of the algorithm.
   - Developed an algorithm to generate all possible matrix minors via index combinations, enabling flexible and scalable matrix analysis.
   - Parallelized minor extraction and determinant computation across distributed nodes using OpenMPI, significantly improving performance over the serial implementation.
   - Set up a distributed computing environment using a master-slave architecture across six servers, utilizing 48 processors and 66 GB of RAM to handle large-scale matrix computations efficiently.
   - **Language/Libraries:** C++, NTL (Number Theory Library), OpenMPI
   - **GitHub repository:** github.com/astroartics/MSc_Project_2025

2. **Development of Numerical Analysis Libraries in C++** — January 2025 - April 2025
   - Designed and implemented numerical analysis libraries from scratch in C++, focusing on clean class design and reusable mathematical functions.
   - Developed core methods for solving systems of linear equations (Gaussian Elimination, Gauss-Jacobi, Gauss-Seidel, LU Decomposition) and implemented matrix operations such as determinant computation, transpose, and multiplication using object-oriented principles.
   - Implemented numerical methods for root finding (Bisection, Newton-Raphson, Fixed-Point) and interpolation (Lagrange's Interpolation, Chebyshev Approximation).
   - Successfully executed matrix computations on 49×49 and 255×255 matrix to test performance and correctness of the implemented algorithms.
   - **GitHub repository:** github.com/astroartics/Numerical_Analysis

Technical skills

Languages: C, C++ | Python (Basic) | Java (Basic)
Developer Tools: Git, Docker, VS Code

Extracurricular activities and achievements

- Ranked 1st in college for all three years of Bachelor's degree (March 2021 - April 2024).
- Completed Level 1 (Certificate) and Level 2 (Diploma) in Spanish at Ranade Institute, SPPU's Department of Foreign Languages (July 2023 - April 2025).
- Secured 1st prize in inter-departmental Chess, Badminton, and Debate competitions (2023–2024).